

MOHAMED SAMIR SALEM

Software Engineering Manager | Applied AI engineer | Full-Stack Developer | Flutter & Backend Architect

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PROFESSIONAL SUMMARY

Software Engineering Manager and Lead Full-Stack Developer with a Master's degree in Computer Programming and 4+ years of experience delivering production-grade mobile, backend, and IoT/AI systems. Top-Rated freelancer on Upwork and founder of two software startups, with a track record of 35+ delivered production systems across e-commerce, healthtech, agritech/IoT, and social platforms. Specializes in Flutter cross-platform development, Java Spring Boot backend architecture, and applied AI/LLM integration (cloud and local/on-prem).

CORE SKILLS

- **Mobile:** Flutter, Dart, BLoC/Cubit, platform channels, native iOS/Android integration, Fastlane, GitHub Actions CI/CD
- **Backend:** Java, Spring Boot, REST/Graph APIs, PostgreSQL, MySQL, SQL/NoSQL, Firebase, Google Cloud, C/C++.
- **Applied AI:** Azure OpenAI, Google Gemini API integration, local LLM orchestration (Ollama, Qwen), Claude code.
- **Practices:** System Design, Clean Architecture, OOD, SOLID, Design Patterns, Code Review, Testing, Agile/Waterfall, Data structure, Algorithms, Git/GitHub, Documentation, Teamwork

WORK EXPERIENCE

Software Engineer | TJAR, Saudi Arabia (Remote)

Jul 2024 – Jul 2025

- Led system design for an e-commerce seller app ([Android/iOS](#)), reducing app size by 22%
- Led DevOps efforts optimizing CI/CD pipelines, cutting deployment time by 30%
- Stack: Flutter, Dart, REST APIs, real-time APIs, cloud messaging, Google Cloud services

Software Engineering Manager / Co-Founder | Intelli / RAI-TECH, Cairo, Egypt *Jul 2022 – Jul 2024 & Aug 2025 - Present*

- Owner of the smart home cross-platform [[Android](#) & [iOS](#)] app and [smart irrigation app](#), [RAI-TECH website](#).
- Owned system design and implementation of a smart home cross-platform app and smart irrigation web app; reduced server cost by 20% and app size by 50%
- Revamped legacy API gateway code, reducing latency by 53%
- Led DevOps team to optimize CI/CD pipelines, cutting deployment time by 45%
- Mentored two interns, reducing operational cost by 12%
- Led on-call operations excellence (OE) rotation as lead
- Stack: Flutter, Dart, Java Spring Boot, real-time APIs, Google Apps Script, cloud messaging, Google Cloud, MySQL

Teaching Assistant | Faculty of Engineering, Helwan University

Dec 2023 – Present

- Deliver course content, facilitate labs, and grade assignments across Problem-Solving, Flutter, and Embedded C tracks
- Coach ICPC Helwan University community heads and instructors
- Contributed to research on adaptive traffic control using deep learning

Software Engineer | EGTAZ, Cairo, Egypt

Dec 2021 – Jul 2022

- Gathered requirements directly with external customers
- Delivered projects on time within an agile team, reducing delivery time by 25%
- Stack: Google Cloud services, Flutter, PHP Laravel, Kotlin, Swift

Software Engineer (Internship) | Samara University Winter Space School, Russia

- Designed a cross-platform Flutter app to calculate rocket design parameters
- Integrated API gateways and real-time APIs for rocket launch archives and schedules

FREELANCE & UPWORK PROJECTS

Top-Rated Freelancer, Upwork | 35+ production systems delivered globally and locally

Lead Mobile Developer / Architect (Startup) | MyGreenPlanet | [Project details](#)

- Built a Flutter social-impact iOS/Android app that turns carbon footprint tracking and offsetting into a shared, gamified experience. It blends environmental utility with social networking to help users build eco-friendly habits.

Backend Architect & Lead Developer (Startup) | Sehetak Bethakaa | [Project details](#)

- AI-driven preventative health platform featuring disease/family history profiling and AI voice/text chat. AI automated medical report scanning, personalized alerts, behavioral commitment tracker, & gamified health quiz leaderboards.

Software Engineer & Architecture Lead | CircleDrop | [Project details](#)

- Built a highly secure fintech and fulfillment platform automating 8-week money group rotations for anonymous item pooling with zero user communication. Engineered a deterministic scheduling matrix alongside a dynamic "Repair Mode" engine to instantly recalculate ledgers if a member misses a payment milestone. Guarded user privacy using zero-knowledge API boundaries while integrating Stripe, retailer inventory tracks, and a real-time risk admin dashboard.

VOLTX website | VOLTX (web infrastructure) | [Project details](#)

- VOLTX is a leading Egyptian manufacturer of high-quality electrical equipment, combining technical expertise with local manufacturing pride. I build Animated, SEO friendly and solid website.

Tag Me Tee | wearable QR contact-sharing | [Project details](#)

- Wearable QR contact sharing on shirt, with ability to manage the redirect url or write a quote.

INDIVIDUAL PROJECTS

Adaptive Ui platform | Liquid glass package— published Flutter package (pub.dev) | [Project details](#)

- A Flutter package that provides adaptive platform-specific widgets with native iOS 26+ designs, traditional Cupertino widgets for older iOS versions, and Material Design for Android.

Full-Stack E-Commerce Platform — Flutter, Java Spring Boot (in development) | [Project details](#)

- stylish and user-friendly e-commerce platform utilizing a third-party API for backend functionality. The backend was developed using Java Spring Boot but is not open source yet.

Smart Home System (Graduation Project) — Flutter, Java Spring Boot, JavaScript, Python, Google Cloud | [Project details](#)

- Wireless smart home devices offer easy installation and cross-platform app control. Features include power and ML-driven voice assistance. My responsibility was to build the app and manage the database and APIs.

Multi-Vendor E-Commerce App — Flutter, Dart, Firebase, Stripe integration | [Project details](#)

- The flutter-based e-commerce app with Firebase backend allows sellers to upload products by category and brand. It enables customers to browse, add items to their cart and wish list, and complete transactions using Stripe payment.

Muslim Guide App (prayer times, Qibla, Tasbih, Quran/Azkar) — Flutter, Dart, Google Apps Script, Kotlin | [Project details](#)

- App to assist with daily habits, including reading the Quran, Azkar, using a digital Tasbih (Sabha), prayer times, & Qibla direction.

Draggable Custom Bottom Navigation Bar — published Flutter package (pub.dev) | [Project details](#)

- Flutter package for a customizable bottom navigation bar, allowing users to add unlimited options as needed.

Flutter Animation Challenges — optimized complex UI animations | [Project details](#)

- Collection of developed and optimized complex animations in Flutter for enhanced user experience and high device performance.

Propeller Display (POV Display) — C/C++, Embedded C (STM) | [Project details](#)

- It enables the presentation of simple animations and video streams; it's controlled by a TV remote control & IR sensor.

PR Robot — Arduino, Embedded C, Python, Embedded Linux | [Project details](#)

- A robot capable of serving as a student activity PR member, engaging with students, responding to inquiries, & distributing flyers

PROBLEM-SOLVING ACHIEVEMENTS

- Codeforces rating 1400+ (top 25% of all users)
- Google Hash Code 2021 — scored 8,234,143 points, top 35% of 8,900+ global teams
- HackerRank Gold: C++, Java, Python, 30 Days of Code, Problem-Solving; Silver: C, SQL

EDUCATION

Master's Degree, Computer Programming — Faculty of Engineering, Helwan University (2023 – 2027)

Bachelor's Degree, Faculty of Engineering, Helwan University — Ranked 1st with Honor (2017 – 2022)